

Orbital Simulator - Technical Document

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Here’s the full LaTeX, also attached above as ‘physics.tex’. The only changes from what you pasted are the four model details. I checked this by comparing it word-for-word with ‘report/chapters/physics.tex’. Besides these four edits, the only differences were the ones already in your pasted version: the emitter paragraph and your shooting-method wording, both kept exactly as you wrote them.

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Chapter 1

Physics

This chapter walks through the physics that the orbital simulator actually implements. The goal is to explain what each part of the model is doing and why, in plain terms, rather than just listing equations. The sections covered here are the electrodynamic tether itself, how the tether is assumed to move, the orbital mechanics that carry the satellite around Earth, the dynamics of the satellite itself including the forces and torques it experiences, and the coordinate frames needed to make all of this consistent.

1.1 Electrodynamic Tether

The payload being modeled is a bare electrodynamic tether. It is a long conducting wire trailed from the satellite, left uninsulated over part of its length so it can make direct contact with the ionospheric plasma. The idea is that a satellite can be deorbited without using any fuel at all, just by dragging a wire through Earth’s magnetic field.

1.1.1 Why current flows at all

As the tether moves through space it also moves through the ionosphere and through Earth’s magnetic field. This motion creates a voltage along the length of the tether, called the motional electric field:

$$E_m = (\mathbf{v}_{rel} \times \mathbf{B}) \cdot \hat{u}_t$$

Here $\mathbf{v}_{rel}$ is the tether’s velocity relative to the surrounding plasma, $\mathbf{B}$ is the local geomagnetic field, computed from the International Geomagnetic Reference Field (IGRF) model evaluated at its 2020 epoch, and $\hat{u}_t$ is the direction the tether points. Wherever the bare part of the wire is positively biased relative to the plasma around it, electrons get collected onto the wire, the same way a charged rod attracts opposite charges. Where the wire is insulated, none of this can happen at all, no matter what the local bias is.

1.1.2 The layout of the tether

The tether is split into two parts. Starting from the exposed tip, there is a bare section of length L_b that can collect current, followed by an insulated section of length L_i that

just carries whatever current has already built up, unchanged, the rest of the way to the spacecraft, where an emitter pushes the collected electrons back out into the plasma. Two conditions pin down the whole problem. At the very tip nothing can flow past the end, so the current there is zero. At the spacecraft end, the voltage is fixed by the emitter.

Two equations govern how the voltage and the current change along the wire. Ohm's law, combined with the motional field, gives

$$\frac{d\varphi}{d\xi} = i(\xi) - 1$$

and orbital motion limited collection theory, the standard model for how a charged cylinder collects current from a plasma, gives, wherever the wire is bare and positively biased,

$$\frac{di}{d\xi} = \frac{3}{4}\sqrt{\varphi}$$

Here everything has been written in a normalized form. φ is the local voltage divided by $|E_m|L^*$, i is the current divided by $|E_m|\sigma_t A_t$, and ξ is the position along the tether divided by a length scale L^* . That length scale is defined as

$$L^* = \left[\frac{(2A_t/p_t)^2 \cdot 9\pi^2 m_e \sigma_t^2 |E_m|}{128 e^3 N_0^2} \right]^{1/3}$$

where A_t and p_t are the tether's cross sectional area and perimeter, σ_t is its conductivity, m_e and e are the electron mass and charge, and N_0 is the ambient plasma density. This length is not the tether's actual length, it is the natural scale over which the wire's own resistance and its ability to pick up current from the plasma balance each other out. Realistic tethers usually work out to a few hundred to a few thousand meters on this scale.

1.1.3 Deriving the closed form solution

Solving the two equations above for a real spacecraft would normally mean numerically integrating a differential equation and adjusting the unknown voltage at the tip until the far end matches the emitter's voltage, which called shooting method. That works and is very accurate, but it takes anywhere from tens of milliseconds to over a second per evaluation, far too slow to call at every step of an orbit propagation. Instead the model uses an exact closed form solution, derived below.

Reducing to one equation

Differentiating Ohm's law and substituting in the collection equation removes the current entirely and leaves a single second order equation for the voltage alone,

$$\varphi'' = \frac{3}{4}\sqrt{\varphi}$$

Multiplying both sides by φ' turns the right hand side into a derivative too,

$$\varphi' \varphi'' = \frac{3}{4}\sqrt{\varphi} \varphi' \implies \frac{d}{d\xi} \left[\frac{1}{2}(\varphi')^2 \right] = \frac{d}{d\xi} \left[\frac{1}{2}\varphi^{3/2} \right]$$

Integrating once gives a first order relation between φ' and φ , with one constant of integration left to fix,

$$(\varphi')^2 = \varphi^{3/2} + C$$

At the tip, $\xi = 0$, the voltage is the unknown φ_A and the current is zero, so $\varphi'(0) = i(0) - 1 = -1$. Plugging this in fixes the constant,

$$C = 1 - \varphi_A^{3/2} \equiv \text{base}$$

so that

$$(\varphi')^2 = \varphi^{3/2} + \text{base}, \quad \varphi' = -\sqrt{\varphi^{3/2} + \text{base}}$$

taking the negative root since the voltage decreases as ξ increases away from the tip.

The hyperbolic substitution

The square root on the right does not have an elementary antiderivative as it stands, but a substitution removes it cleanly. Let

$$\varphi = \text{base}^{2/3} \sinh^4(v)$$

for an auxiliary variable v . Then $\varphi^{3/2} = \text{base} \sinh^2(v)$, so

$$\varphi^{3/2} + \text{base} = \text{base} (\sinh^2(v) + 1) = \text{base} \cosh^2(v)$$

using the identity $\cosh^2(v) - \sinh^2(v) = 1$. The square root collapses to $\sqrt{\text{base}} \cosh(v)$, so

$$\varphi' = -\sqrt{\text{base}} \cosh(v)$$

and since $\varphi' = i - 1$, the current comes out directly in terms of v ,

$$i = 1 - \sqrt{\text{base}} \cosh(v)$$

Turning the substitution into a position along the tether

Differentiating the substitution and combining it with the separable equation above gives, after simplifying,

$$d\xi = -\frac{4}{3} \text{base}^{1/6} \sinh^{1/3}(v) dv$$

Integrating this out to some value of v requires the integral of $\sinh^{1/3}$, which has no elementary closed form. This integral is defined once and for all as

$$F_s(z) = \int_0^z \sinh^{1/3}(\zeta) d\zeta$$

and computed a single time as a lookup table when the program starts, so that afterward any value of F_s , or its inverse, is available instantly rather than being integrated again. With that definition,

$$\xi(v) = \frac{4}{3} \text{base}^{1/6} \left(F_s(v_{c0}) - F_s(v) \right)$$

where v_{c0} is the value of v matching the tip's known voltage φ_A , found from the substitution above and equivalently written as

$$\cosh(v_{c0}) = \frac{1}{\sqrt{\text{base}}}$$

The crossing point and the frozen region

The voltage reaches zero exactly where $v = 0$, since $\sinh(0) = 0$. Setting $v = 0$ in the position formula gives the crossing point,

$$\xi_{AB} = \frac{4}{3} \text{base}^{1/6} F_s(v_{c0})$$

This is not chosen in advance, it falls out of the solution itself. Two cases follow depending on how this compares to the bare length ξ_b .

If the bare segment ends before reaching the crossing, $\xi_b \leq \xi_{AB}$, then the value of v at the end of the bare segment, call it v_b , is found by inverting the position formula,

$$F_s(v_b) = F_s(v_{c0}) - \frac{3\xi_b}{4\text{base}^{1/6}}$$

and the voltage and current at that point follow directly from the substitution,

$$\varphi_P = \text{base}^{2/3} \sinh^{4/3}(v_b), \quad i_P = 1 - \sqrt{\text{base}} \cosh(v_b)$$

If instead the bare segment is long enough to reach the crossing, $\xi_b > \xi_{AB}$, collection effectively stops there and the small remaining ion current is treated as negligible, the same simplification used in the reference model this is built from. The current is then frozen at whatever value it reached at the crossing,

$$i_B = 1 - \sqrt{\text{base}}$$

and the voltage simply continues in a straight line for the rest of the bare segment,

$$\varphi_P = (i_B - 1)(\xi_b - \xi_{AB})$$

The insulated segment and closing the loop

Once the wire is insulated no more current can be collected no matter what the local voltage is, so the current stays fixed at whatever value i_P it reached at the end of the bare segment. Ohm's law is still valid though, so the voltage keeps changing at a constant rate for the rest of the way to the spacecraft,

$$\varphi_{Si} = \varphi_P + (i_P - 1) \xi_i$$

The only remaining unknown in the whole problem is φ_A , the voltage at the tip, and it is found by adjusting it until this final expression matches the voltage fixed by the emitter at the spacecraft, φ_C . This is a single equation in a single unknown, solved with a standard one dimensional root finder, needing only a handful of evaluations of the closed form expressions above rather than any integration of a differential equation. The whole approach runs in about a tenth of a millisecond, roughly ten thousand times faster than solving the differential equation directly, while matching it to eight significant figures.

The special case at saturation

As φ_A approaches its maximum possible value of one, base approaches zero, and several of the expressions above involve dividing by base or by $\sqrt{\text{base}}$, which becomes numerically unstable. Right at this limit an exact, simpler closed form is used instead,

$$\varphi(\xi) = \left(1 - \frac{\xi}{4}\right)^4, \quad i(\xi) = 1 - \left(1 - \frac{\xi}{4}\right)^3$$

valid up to $\xi = 4$, beyond which the voltage stays at zero and the current stays at its maximum value of one. This also explains why the crossing point can never exceed four tether length scales, since that is the largest it can ever grow to reach.

1.1.4 The number that matters for the force

For computing the force on the tether, what is actually needed is not the whole current profile along the wire, just its average value. Since Ohm's law says $\varphi' = i - 1$ everywhere along the wire, whether it is bare or insulated, integrating both sides from the tip to the spacecraft end and using the fundamental theorem of calculus gives

$$\int_0^{\xi_a} i d\xi = \xi_a + \int_0^{\xi_a} \varphi' d\xi = \xi_a + (\varphi_C - \varphi_A)$$

Dividing by the total normalized length ξ_a gives the average current directly,

$$i_{av} = \frac{1}{\xi_a} \int_0^{\xi_a} i d\xi = 1 - \frac{\varphi_A - \varphi_C}{\xi_a}$$

This is not an approximation. It comes directly out of integrating $\varphi' = i - 1$ across the whole tether, so it gives the exact average current with no need to sample the current profile at many points and add it up numerically.

1.1.5 What this leaves out

The model only covers the standard passive mode of a bare tether, where current flows on its own with no power supply choosing its direction, and it treats the ion collection that happens in the tiny region past the electron collecting zone as negligible, which is the same simplification used in the reference literature this model is built from. There is also the case, covered above, where the tip voltage saturates at its maximum possible value, where the general formula becomes numerically unstable and a separate exact formula is used instead. Finally, the ambient plasma density N_0 is currently a placeholder constant rather than a real, changing value with location and time, which is the single biggest source of uncertainty in the actual current numbers this model produces, even though the shape of the physics is solid.

The major thing that we did not consider yet is the voltage drop created by the special type of emitter instead of an assumed value. So, depending on the emitter type and how it functions, we get a voltage drop at the tether end. This is an important gap that needs to be filled.

1.2 Tether Dynamics

Dynamics here means how the tether actually moves and points, as opposed to the electrical physics of the previous section.

1.2.1 The tether is assumed to always point straight down

At every instant the tether's direction is recomputed as pointing from the satellite straight toward the center of Earth, the direction usually called nadir. This is not an arbitrary choice. Near the equator Earth's magnetic field runs mostly horizontally, roughly north to south, and the satellite's orbital velocity is also roughly horizontal, along the direction of travel. The cross product of two horizontal, roughly perpendicular vectors points close to vertical, so a tether hanging vertically happens to line up with the direction where the induced voltage is naturally the largest. This is exactly why real gravity gradient stabilized tethers are built to hang vertically in the first place.

1.2.2 What is not modeled

The tether is treated as a perfectly straight, perfectly rigid rod with no give at all. In reality a tether behaves more like a pendulum. Gravity gradient forces pull it back toward vertical, but they do not hold it perfectly still, and real tethers swing, both in the plane of the orbit and out of it. There is even a well known area of research into how a tether's own current can drive this swinging into an unstable, growing oscillation rather than a gentle one. None of that swinging motion exists in this simulator. The tether is simply always exactly vertical, at every timestep, regardless of anything else going on.

1.2.3 The satellite's own rotation is separate

The satellite body has its own attitude simulation, propagated using the standard equations of rigid body rotation under an applied torque. As things stand, no torque is ever actually applied. The attitude control subsystem is a placeholder that always outputs zero torque, and no other torque source, whether from gravity gradients, aerodynamic drag, or the tether's own force, feeds into the attitude equations either. With the default spacecraft inertia, which is symmetric, this means the satellite simply spins at a constant rate about a fixed axis with no wobble at all, since the mathematics works out so that a symmetric body under zero torque cannot wobble. With an uneven inertia the satellite could in principle tumble in a complicated way even with no torque applied, purely from its own mass distribution, but that tumbling, if it happened, would still have no connection to the tether's assumed direction. The tether's orientation and the satellite body's rotation are two completely separate calculations in this model, and nothing links them together.

1.3 Orbital Mechanics

1.3.1 Describing the orbit

An orbit is described using six numbers, the classical orbital elements. Two describe its size and shape, the semi major axis and the eccentricity. Three describe how the orbital plane is tilted and oriented in space: the inclination, which is how tilted the plane is relative to the equator, the right ascension of the ascending node, often shortened to RAAN, which is where around the equator the orbit crosses from south to north, and the argument of periapsis, which describes the orientation of the orbit's shape within that plane. The sixth number says where along the orbit the satellite currently sits. In this simulator these six numbers are set once, at the very start, and the orbit is then handed off to a numerical integrator that carries it forward in time.

Before integration begins, the orbit is converted from these classical elements into an equivalent set called equinoctial elements. The classical elements have a mathematical problem, some of them stop being well defined for a circular or perfectly flat orbit, since terms like where the closest approach is start dividing by something close to zero. Equinoctial elements describe exactly the same physical orbit without that problem, so the numerical integrator is given a representation that behaves well no matter how close to circular the orbit is.

1.3.2 Carrying the orbit forward in time

The actual propagation uses an adaptive step size integrator, specifically an eighth order Runge Kutta method, the Dormand–Prince 8(5,3) integrator, running inside Orekit's numerical propagator with a minimum step of 0.001 s, a maximum step of 500 s, and error tolerances set for roughly one meter of position accuracy. Adaptive means the integrator does not take fixed size steps in time. It estimates its own error at each step, and takes larger steps when the motion is smooth and smaller ones when things are changing quickly, always staying within chosen minimum and maximum step sizes. This matters for understanding how often each force actually gets recalculated. Over long stretches where nothing dramatic is happening, the integrator can go for several minutes of simulated time between calls to a given force model, even though results are reported to the user on a much finer, fixed time grid.

Gravity is not modeled as a simple point mass pulling on the satellite. Earth is not a perfect sphere and its mass is not evenly spread out, so the model uses a spherical harmonics expansion of the real gravity field, the Holmes–Featherstone formulation using the EIGEN-6S gravity field coefficients, carried out to degree and order 20, which captures the equatorial bulge and other larger scale lumps in Earth's gravity.

1.3.3 Slow changes over many orbits

Small forces acting continuously, like drag and the tether's own force, do not noticeably bend a single orbit, but they add up. Over many orbits the semi major axis slowly shrinks, which is the deorbiting effect the whole tether is there to produce. At the same time, the orientation

of the orbital plane itself slowly rotates, showing up as a steady drift in the inclination and the RAAN. Most of that drift, for a typical low orbit, comes from Earth’s equatorial bulge and would happen with no tether at all. The tether adds its own smaller contribution on top of that, sometimes helping the natural drift along and sometimes working against it, depending on the exact orbit.

1.4 Satellite Dynamics

This section covers how the satellite itself is pushed and turned: the forces that change its orbit, the torques that would change its orientation, and how the rotational motion is actually solved numerically.

1.4.1 Forces

Three separate forces are applied to the satellite in this simulator, and they are simply added together at every step of the integration. Nothing is treated specially or subtracted, each force model hands over a vector and the propagator sums them all before deciding how the orbit changes.

Gravity, described in the previous section, is by far the largest of the three. Atmospheric drag comes from the Harris–Priester model, a standard empirical model of the upper atmosphere’s density, which also accounts for the atmosphere being denser on the side of Earth facing the sun, combined with the satellite’s cross sectional area and a drag coefficient, treated as isotropic so that the drag area does not depend on the satellite’s orientation. Drag always acts opposite to the satellite’s velocity relative to the atmosphere, which is assumed to rotate along with Earth.

The tether force uses the textbook formula for the force on a current carrying wire in a magnetic field,

$$\mathbf{F} = I_{av} L (\hat{u}_t \times \mathbf{B})$$

using the total tether length L and the average current I_{av} derived in the electrodynamic tether section, converted back into physical units of amps. A natural question is whether using an average current here is an approximation, since the real current is not the same everywhere along the wire. It turns out it is not an approximation at all. Integrating the true, varying current along the wire and multiplying the average by the full length give exactly the same total force, because of how the average current and the length scale L^* are defined in the first place. The two match exactly, not just approximately.

A passive tether is only useful if the current that develops on its own happens to always produce a braking force, never a boost. Whether that holds comes down to a single detail, whether the sign of the current correctly follows the sign of the motional voltage E_m described earlier. When it does, a short calculation using a standard vector identity shows that the force dotted with the satellite’s velocity relative to the plasma always comes out negative or zero, which means the tether force can never do positive work on the orbit. It is always some form of drag, never thrust, as long as the current’s direction is computed correctly from E_m . This matches how real passive bare tethers are described in the literature, where switching

the sign of the current is exactly what turns the same hardware from a drag device into a thruster, and a passive tether left alone always ends up on the drag side.

1.4.2 Geometry and mass properties

The satellite's physical shape and its mass properties are handled as two entirely separate things in this model, with nothing connecting them.

Mass properties, meaning the total mass, the drag cross sectional area, the drag coefficient, and the three principal moments of inertia discussed below, are each entered independently as flat numbers. None of them are computed from an actual shape or a real distribution of mass inside the satellite. It is left entirely up to whoever fills in these numbers to make sure they are consistent with each other and with some real, physical satellite design.

Separately, a three dimensional mesh file describing the satellite's outer shape is loaded purely for the attitude display, so that a small model can be shown spinning on screen in a way that matches the propagated orientation. This mesh has no effect on anything else in the simulation. Its actual shape is never used to work out how much cross sectional area faces the direction of travel as the satellite turns, and it is never used to compute the moment of inertia tensor from the satellite's true geometry and mass distribution. The visual model and the physical numbers used in the simulation are two disconnected representations of the same satellite, and nothing checks that they agree with one another.

1.4.3 Torques

Where forces change the satellite's position and velocity, torques change how it is oriented and how fast it is spinning. A real spacecraft can experience several kinds of torque at once. Gravity gradient torque comes from the fact that the part of the satellite closer to Earth feels a very slightly stronger pull than the part farther away, which twists the satellite toward alignment with the local vertical. Aerodynamic torque comes from drag acting off center if the satellite's shape is not symmetric about its center of mass. There can also be torque from the magnetic field acting on any residual magnetization of the spacecraft, from solar radiation pressure pushing unevenly on different surfaces, and from onboard actuators such as reaction wheels or magnetorquers deliberately commanding a torque to point the satellite a certain way.

In this simulator, none of the environmental torques above are modeled. The only torque input the rotational equations accept comes from the attitude control subsystem, and that subsystem is currently a placeholder that always outputs zero. There is also no torque computed from the tether force itself, even though a real tether pulling on the spacecraft away from its center of mass would generically produce one. As it stands, the satellite experiences no torque of any kind from any source.

1.4.4 Rigid body dynamics implementation

The satellite's orientation is tracked using a quaternion q , a four number representation of a rotation that avoids the singularities that roll, pitch, and yaw angles run into, together

with an angular velocity vector $\boldsymbol{\omega}$ in the satellite's own body frame. Two equations advance these forward in time. The quaternion changes according to

$$\dot{q} = \frac{1}{2} \Omega(\boldsymbol{\omega}) q$$

where $\Omega(\boldsymbol{\omega})$ is a matrix built from the angular velocity components, and the angular velocity itself changes according to Euler's rigid body equation,

$$\dot{\boldsymbol{\omega}} = I^{-1} \left(\boldsymbol{\tau} - \boldsymbol{\omega} \times (I \boldsymbol{\omega}) \right)$$

where I is the satellite's moment of inertia tensor and $\boldsymbol{\tau}$ is the applied torque. The term $\boldsymbol{\omega} \times (I \boldsymbol{\omega})$ is not an external effect at all, it is the natural coupling between spin axes that any rotating body with unequal moments of inertia experiences, present even with zero applied torque, and it is what makes an object like a tumbling wrench or a thrown football wobble in a way that looks complicated even though nothing is pushing on it.

In the implementation, I is built as a diagonal matrix from three separately configurable numbers, the principal moments of inertia about the body's own x, y, and z axes. Building it this way assumes those three body axes already line up exactly with the satellite's true principal axes, so there are no off diagonal cross terms, sometimes called products of inertia, anywhere in the tensor. For a real, irregularly shaped satellite this is an approximation, since the true principal axes generally do not line up perfectly with whatever axes are convenient to define on the structure, but it is a common and reasonable simplification for an early stage model. The default configuration sets all three moments equal to one another, which makes the satellite perfectly symmetric and is exactly why, as described below, it does not wobble at all under the default settings.

Both equations are advanced together at each timestep using a fourth order Runge Kutta integration, a standard numerical scheme that evaluates the equations of motion four times per step at carefully chosen intermediate points and combines them for a result accurate to fourth order in the step size.

With torque always at zero, the second equation reduces to $\dot{\boldsymbol{\omega}} = -I^{-1}(\boldsymbol{\omega} \times (I \boldsymbol{\omega}))$, motion driven purely by the satellite's own inertia. If the three moments of inertia are equal, meaning the satellite is modeled as perfectly symmetric, then $I \boldsymbol{\omega}$ points in exactly the same direction as $\boldsymbol{\omega}$, and the cross product of any vector with something pointing the same way is exactly zero. In that case the angular velocity never changes at all, and the satellite simply spins at a constant rate forever. With an uneven inertia this term does not vanish, and the satellite can tumble in a complicated way with no torque applied at all, since a spinning body with three different moments of inertia does not conserve its spin axis, only its total angular momentum.

1.4.5 What needs to be added

A number of pieces are either missing or simplified in the current implementation of satellite dynamics.

- A working attitude control law. The attitude control subsystem currently outputs zero torque at all times, with no pointing control, no detumbling, and no slew logic implemented.

- Environmental torques. Gravity gradient torque, aerodynamic torque, and any torque from residual magnetization are not modeled at all, even though gravity gradient torque in particular is usually the dominant natural torque for a satellite at these altitudes.
- A torque from the tether force. The tether force is currently applied only as a translational push on the satellite's center of mass. A real tether pulling from a point away from the center of mass would also produce a torque on the spacecraft body, which is not computed anywhere in this model.
- The tether's own dynamics. The tether is currently treated as an idealized, perfectly rigid, always vertical line with no ability to swing or respond to forces on its own, as described in the tether dynamics section above. A more complete treatment would model the tether as its own dynamical system, including its libration around the local vertical and its own rigid or flexible body behavior, rather than assuming its orientation outright.
- Mass properties computed from geometry. The moment of inertia tensor and the drag cross sectional area are currently independent, hand entered numbers rather than values computed from the satellite's actual three dimensional shape and mass distribution. Deriving them directly from a real geometric model, the same one already used for the visual display, would remove the risk of these numbers being inconsistent with each other or with the physical design they are meant to represent.
- The Tether and Satellite are disconnected in terms of torsional dynamics.

1.5 Frames

1.5.1 Why more than one coordinate system is needed

Different physical quantities are naturally described in different coordinate systems, and a few conversions are unavoidable before they can be combined. The satellite's motion is simplest to describe in an inertial frame, one that does not rotate with Earth, since Newton's laws only take their simple form when the frame itself is not spinning. Latitude, longitude, and the strength of the magnetic field only make sense relative to Earth's surface, so they are naturally described in a frame that rotates along with the planet. A magnetic field model reports its answer relative to compass directions at one specific point on the ground, which is yet another frame, one that only exists locally and points in a different absolute direction depending on where on Earth you are standing.

None of these conversions change what a vector physically is. Converting a vector from one frame to another is exactly like translating a sentence into another language, the underlying meaning stays the same, only the words used to describe it change. A vector is really just three numbers relative to a chosen set of axes, and the same physical arrow has different numbers depending on which axes are chosen.

1.5.2 The frames actually used

The following frames appear in the model.

- An inertial frame, which does not rotate, and is what the orbit propagator itself works in.
- An Earth fixed frame, which rotates once a day along with the planet, and is where latitude, longitude, and the magnetic field are meaningful.
- A geodetic description, latitude, longitude, and altitude above a reference ellipsoid, which is not really a set of axes at all but a location description derived from the Earth fixed position.
- A local north, east, down frame, which is the convention the magnetic field model reports its answer in, and only makes sense at one specific point on the ground.
- A local east, north, up frame, which carries the same information as north, east, down but is the convention the simulator's own rotation tools expect.
- A local vertical, local horizontal frame, tied to the satellite's position in its orbit rather than a point on the ground, used for describing the satellite's attitude relative to its direction of travel.

1.5.3 The chain used for the tether force

Computing the tether force in particular needs the following sequence of conversions, repeated at every evaluation. Start with the satellite's position and velocity in the inertial frame, since that is what the propagator provides. Rotate into the Earth fixed frame, since that is where the magnetic field and the satellite's motion relative to the plasma both need to be evaluated. Convert the Earth fixed position into latitude, longitude, and altitude, since the magnetic field model needs that as its input. Look up the magnetic field, which comes back in the local north, east, down convention. Relabel those same three numbers into east, north, up, which is not a rotation at all, just a reordering and a sign change on one component. Rotate that local description into the global Earth fixed frame, so it can finally be combined with the satellite's own position and velocity, which are already in that frame. Compute the actual force there. Then rotate the result back into the inertial frame, since that is the one frame the orbit propagator is allowed to accept a force in.

Every one of these steps except the actual force calculation is bookkeeping. The physical magnetic field, and the physical force it produces, do not change at all during this chain. Only the numbers used to describe them change, so that by the time two vectors need to be combined, they are finally expressed using the same set of axes. “